Note on Time Crystal

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Abstract

A brief note on time crystal.

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I. Failure of continuous time crystal

Starting from Wilczek[1], the concept of time crystal comes into our sight, for one possible aim to fill the scheme of spontaneous symmetry breaking in time domain. The original idea is to consider a system with a time independent Hamiltonian in equilibrium, but with its certain chosen ground state breaking the continuous time translational symmetry. In the proposal, the Hamiltonian is composed of a flux threaded 1D Bosoinc ring of unit length with attractive point interaction,

$$H = \sum_{i=1}^{N} \frac{(p_i - A)^2}{2} + \frac{g_0}{2} \sum_{i \neq j} \delta(x_i - x_j), \tag{1}$$

with a simplified unit choice. When $A = 0$, the Hamiltonian shares a bright soliton solution around the center-of-mass at the mean-field level forming Bose-Einstein condensate for strong-enough attraction g_0, and Wilczek expected that when $A \neq 0$ as it is turned on adiabatically, a periodic motion of the soliton will be observed, which is the time crystal. However, in the center-of-mass frame[2], the center decouples with the relative motion as

$$H = \frac{(P - NA)^2}{2N} + \cdots, \tag{2}$$

and the ground state requires that

$$0 = \frac{\partial H}{\partial P} = 2\pi \frac{j}{N} - \alpha, \tag{3}$$

which can be exactly fulfilled when $N \to \infty$, with $g_0(N-1)$ setting as a constant. Notice that $j \in \mathbb{Z}$ denotes the quantum number for momentum. The result illustrates that there is no probability current, which is the same as the right-hand side, related to the center of mass degree of freedom, and the proposed picture is invalid. Interestingly, Syrwid et al. showed that the excited states of the system can exhibit time crystal behavior[2], by setting a much higher j such that the probability current is non-zero.

More generally, Oshikawa et al.[3] proved that a continuous time crystal — which breaks the continuous time translation symmetry spontaneously — is impossible for the ground state of an equilibrium system. Their proof is very robust and forms a no-go theorem based on simply a local Hamiltonian assumption. They also provide an accurate definition for what does it mean when an order forms in the time domain. Let us follow their definitions. A spatial long-range order means that the equal-time correlation function of a local order-parameter $\phi(x,t)$ satisfies that

$$\lim_{V\to\infty} \langle \phi(x,t)\phi(x',t)\rangle \to \sigma^2 \neq 0, \tag{4}$$

for $|x-x'|$ much greater than any microscopic scales. Notice that for a crystalline order, the right-hand side should be periodic with $x-x'$, i.e., $f(x,x') = f(x-x')$ with $f \neq const$, or equivalently we can write its Fourier component as $f_{G\neq 0} \neq 0$, which signals a density wave. The expectation value is defined over canonical or grand-canonical ensemble. Notice that when we focus on the zero-temperature limit, this expectation can be evaluated over *any* ground state, i.e., even if a system holds a ground state with spontaneously broken symmetry, this correlation can not tell over which ground state it is evaluated. This can be either a ground state with broken symmetry or a ground state with unbroken symmetry. Let us keep this point in mind and push it to the time domain, and correspondingly we write in the Heisenberg picture that

$$\lim_{V\to\infty} \langle \phi(x,t)\phi(0,0)\rangle \to f(t), \tag{5}$$

which should be independent of x and exhibit non-trivial periodic oscillation in t. Essentially notice that we can not write an equal-position correlation function, since this can not exclude

some trivial cases. By letting $\Phi(t) = \int_V \mathrm{d}x\phi(x,t)$, we can further write

$$\frac{1}{V^2}\lim_{V\to\infty}\langle\Phi(t)\Phi(0)\rangle = f(t), \tag{6}$$

where we have ignored the possibility of co-existence with spatial crystalline order to form a space-time crystal. And in this sense, Oshikawa et al. proved that when evaluating Eq. (6) in equilibrium for a system with locality, the right-hand side is always a constant, which means that continuous time crystal is impossible. The proof is divided into two parts, for zero-temperature and finite-temperature cases, although the latter is general.

($T = 0$.) The aim is to prove that the correlation is constant in time, so we can consider its change in time as ($\Phi = \Phi(0)$)

$$\begin{aligned}\langle 0|\Phi(t)\Phi|0\rangle - \langle 0|\Phi\Phi|0\rangle &= \int_0^t \mathrm{d}s\frac{\mathrm{d}}{\mathrm{d}s}\langle 0|\Phi e^{-i(H-E_0)s}\Phi|0\rangle \\ &= -i\int_0^t \mathrm{d}s\,\langle 0|\Phi(H-E_0)e^{-i(H-E_0)s}\Phi|0\rangle\,,\end{aligned} \tag{7}$$

where $H - E_0$ is positive definite. Consider generally, and with the help of Schwarz inequality $\langle AB\rangle \le \sqrt{\langle AA^\dagger\rangle\langle BB^\dagger\rangle}$, we have

$$\begin{aligned}\langle 0|A(H-E_0)e^{-i(H-E_0)s}B|0\rangle &\le \sqrt{\langle 0|A(H-E_0)A|0\rangle\langle 0|B(H-E_0)B|0\rangle} \\ &= \frac{1}{2}\sqrt{\langle 0|[A,[H,A]]|0\rangle\langle 0|[B,[H,B]]|0\rangle}.\end{aligned} \tag{8}$$

Now consider that A, B are integrals of 0-form local order parameters, and by assumption H is also local, we have $O(A) = O(B) = O(H) = O(V)$, and

$$[H,A] = \int \mathrm{d}x\mathrm{d}y[h(x),a(y)] = \int \mathrm{d}x[h(x),a(x)] = O(V), \tag{9}$$

thus $[A,[H,A]] = O(V^3/V^2) = O(V)$, and we can write

$$|\,\langle 0|\Phi(t)\Phi|0\rangle - \langle 0|\Phi\Phi|0\rangle\,| \le \int_0^t \mathrm{d}s CV = CtV, \tag{10}$$

with C some constant. Following the definition of long-range order, we have

$$|f(t) - \langle 0|\Phi\Phi|0\rangle/V^2| \le \lim_{V\to\infty} C\frac{t}{V} = 0, \tag{11}$$

which means that $f(t)$ is a constant.

($T \neq 0$.) A direct extension of the above proof fails since $H - E_n$ is not positive definite generally. Nevertheless, a similar consideration concerning whether the correlation function

evolves in time can be addressed with the help of spectral decomposition and the Lieb-Robinson bound[4] for a local Hamiltonian

$$\Big\|[a(x,t),b(0,0)]\Big\| \leq \min\{C_1, C_2 e^{-\mu(|x|-vt)}\}, \tag{12}$$

where the constants does not depend on x, t, and the bound reflects a finite information propagation speed. By definition, we can decompose

$$\begin{aligned} f_{AB}(t) &= \frac{1}{V^2}\langle A(t)B(0)\rangle \\ &= \frac{1}{ZV^2}\sum_{n,m} e^{-\beta E_n}\langle n|A|m\rangle\langle m|B|n\rangle\, e^{i(E_n-E_m)t} \\ &= \int \mathrm{d}\omega e^{-i\omega t}\rho_{AB}(\omega), \end{aligned} \tag{13}$$

where Z is the partition function, and $\rho_{AB}(\omega)$ is the spectral function that reads

$$\rho_{AB}(\omega) = \frac{1}{ZV^2}\sum_{n,m} e^{-\beta E_n}\langle n|A|m\rangle\langle m|B|n\rangle\,\delta(\omega - E_m + E_n), \tag{14}$$

and if we can prove that $\rho_{AB}(\omega \neq 0) = 0$, then $f_{AB}(t)$ is a constant. To achieve this, we utilize the Lieb-Robinson bound to write

$$\begin{aligned} g_{AB}(t) &= \frac{1}{V^2}\langle [A(t),B(0)]\rangle \\ &= \frac{1}{V^2}\int \mathrm{d}x\mathrm{d}y\,\langle [a(x,t),b(y,0)]\rangle \\ &= \frac{1}{V}\int \mathrm{d}x\,\langle [a(x,t),b(0,0)]\rangle \to 0, \end{aligned} \tag{15}$$

where in the last step we used the bound relation which tells that the integral has to be finite at any finite t. On the other hand, we can write

$$\begin{aligned} g_{AB}(t) &= f_{AB}(t) - f_{BA}(-t) \\ &= \int \mathrm{d}\omega e^{-i\omega t}\rho_{AB}(\omega) - \int \mathrm{d}\omega e^{i\omega t}\rho_{BA}(\omega) \\ &= \int \mathrm{d}\omega e^{-i\omega t}[\rho_{AB}(\omega) - \rho_{BA}(-\omega)] \\ &= \int \mathrm{d}\omega e^{-i\omega t}\rho_{AB}(\omega)[1 - e^{-\beta\omega}], \end{aligned} \tag{16}$$

and since $g_{AB}(t) = 0$ for all t, we have $\rho_{AB}(\omega)[1 - e^{-\beta\omega}] = 0$, thus $\rho_{AB}(\omega) = 0$ for $\omega \neq 0$. This means that $f_{AB}(t)$ is a constant, and by letting $A = B = \Phi$, we have proved that the continuous time crystal is impossible for an equilibrium system.

Indeed, based on a simple effective Lagraingian argument, Zaletel et al.[5] showed that a continuous time crystal is impossible for a quantum system in equilibrium. Generalizing the Ginzburg-Landau theory to the time domain, we can write the effective Lagrangian as

$$\mathcal{L}_k = c_4|\partial_t\phi|^4 + c_2|\partial_t\phi|^2, \tag{17}$$

with $c_4 > 0, c_2 < 0$ revealing a spontaneous broken time translation symmetry as $\langle\partial_t\phi\rangle \neq 0$. However, in equilibrium, minimization of the free energy leads to the Hamilton's equation which reads that $\partial_t\phi = \partial_p H = 0$ with $p = \partial\mathcal{L}_k/\partial(\partial_t\phi)$ the conjugate momentum, and a ground state cannot exhibit even simple order parameter evolution, not to mention periodic oscillations. The seeming possibility for spontaneous symmetry breaking lies in the non-single-valued nature of $\partial_t\phi$ as a inverse function of p, which is not the case for a physical system.

II. Rise of discrete time crystal

We have saw that a naïve continuous time crystal is impossible for an equilibrium system, simply due to the thermodynamic limitation for a local Hamiltonian. This indeed reveals that time and space are not quite equivalent, at least when considering the non-relativistic condensed matter systems, possibly due to the low-energy nature and the tendency for the emergence of a certain time-direction arrow. Nevertheless, the idea of a time crystal is not completely killed, while now we can not rely on a static Hamiltonian — it needs to be either driven or open[5]. Here, we shall focus on the first case, and consider a periodically driven system known as a Floquet Hamiltonian. In such systems, the Hamiltonian is time-periodic, i.e., $H(t) = H(t + T)$, with T the period indicating a discrete time-translation symmetry, which can be spontaneously broken signaled by a subharmonic response, indicating a *discrete time crystal* (DTC). Under the scheme, two possibilities have been proposed. The first one follows possibly from the Kolmogorov-Arnold-Moser (KAM) theorem that applies in classical systems and proves that quasiperiodic orbits of dynamical systems remain robust to small perturbations, and the corresponding Floquet version states that the dynamics can exhibit Floquet prethermalization in the limit of large driving frequencies, with a discrete time-crystalline order that survives with an exponentially increasing lifetime with the frequency of the drive. The other one is more likely a stable 'phase' where the time crystalline order

survive permanently in the thermodynamic limit, and to realize this both strong disorder and many-body interactions are needed, creating many-body localization that breaks ergodicity and protects the time-crystalline order.

A. Ordinary spontaneous symmetry breaking

Before diving into the realm of DTC, it is worthy to give more discussions on in what sense spontaneous symmetry breaking happens[6, 7]. Admittedly, in the high temperature limit, an isolated time-independent system in equilibrium is always in a disordered phase, and the symmetry is unbroken. Then a first claim is that a phase transition for a finite-size system to cool adiabatically to an ordered phase is impossible, since the partition function for such a system is always analytic in the temperature, so does the order parameter. In other words, (for the non-degenerate case) the eigenstates of the finite-size Hamiltonian are always at the same time eigenstates of the symmetry operator, thus eliminating the possibility to contain an symmetry-breaking order. In this sense, we have to work in the thermodynamic limit, where N/V is fixed although both go to infinity, to see the spontaneous establishment of an order through a phase transition promised by the Lee-Yang theory. Basically, such a process can be expressed as an unexchangeable singular limit as

$$\begin{cases} \lim_{N\to\infty} \lim_{h\to 0} O = 0 \\ \lim_{h\to 0} \lim_{N\to\infty} O \neq 0, \end{cases} \tag{18}$$

where O is a measurable order and h is the related external field, and in the second line the symmetry breaking happens with the limit value of O depending on the manner h goes to zero, indicating the spontaneous nature. Assuming that we are now in the zero temperature limit, and the established order pushes us to define an order parameter operator, whose expectation value, namely the order parameter, can provide us observable quantities and is a local one in our simple case. The order parameter operator does not necessarily commute with the Hamiltonian for a finite-size system, although the commutator expectation value vanishes in the thermodynamic limit. The order parameter operator should not commute with the symmetry operator, as its eigenstates are considered to be symmetry-broken states, and inversely, the symmetric states should have vanishing order parameter. Furthermore, a good order parameter operator should be able to distinguish as accurate as possible the

symmetry-broken space, reflected by the different non-vanishing eigenvalues of the order parameter for the ground states of the Hamiltonian.

A formal expression for the spontaneous symmetry breaking process lies in reducing the global symmetry group G of the Hamiltonian into its subgroup H, with the coset G/H counting the upper-bound number of the Goldstone modes. For instance, the scaler Higgs Boson spontaneously breaks of the $U(1)$ symmetry into the trivial group, with one Goldstone mode further coupled with the gauge field to make the Goldstone mode massive. Formally, let $U = e^{i\alpha Q} \in G$ with Q the generator of the symmetry transformation, then a state $|\psi\rangle$ breaks the symmetry if there exists a locally defined *interpolating field operator* $I(x)$ such that

$$\langle\psi| \left[Q, I(x)\right] |\psi\rangle \neq 0. \tag{19}$$

The definition is most easily understood by considering a symmetric state $|\psi\rangle_{Sym}$, which makes the expectation value to vanish automatically, since eigenvalue of $|\psi\rangle_{Sym}$ under a Hermitian Q has to be real. The local order parameter operator can be defined as $\phi(x) = [Q, I(x)]$, which is not necessarily Hermitian. To be complete, let us pay attention to the Goldstone mode as the gapless excitations accompanied with the broken global continuous symmetry. Consider the complete spectrum of Hamiltonian $H|n,k\rangle = E_n(k)|n,k\rangle$, with orthonormal relation $\langle n,k|n',k'\rangle = (2\pi)^d\delta_{nn'}\delta(k-k')$ and complete relation $\mathbb{I} = \sum_n \int \frac{\mathrm{d}^d k}{(2\pi)^d} |n,k\rangle\langle n,k|$ in $D = d+1$ spacetime. Being a symmetry, the Noether charge Q corresponds to a local current conservation formula $\partial_\mu j^\mu = 0$, with $Q(t) = \int \mathrm{d}^d x j^0(x,t)$ satisfying that $\partial_t Q = 0$. Then for the symmetry breaking ground state,

$$\begin{aligned}
0 \neq \langle GS|\left[Q, I(y)\right]|GS\rangle &= \int \mathrm{d}^d x \left(\langle GS|j^0(x,t)I(y)|GS\rangle - \text{c.c.}\right) \\
&= \int_\Omega \mathrm{d}^d x \sum_n \int \frac{\mathrm{d}^d k}{(2\pi)^d} \left(\langle GS|e^{i(Ht-p\cdot x)}j^0(0)e^{-i(Ht-p\cdot x)}|nk\rangle\langle nk|I(y)|GS\rangle - \text{c.c.}\right) \\
&= \int_\Omega \mathrm{d}^d x \sum_n \int \frac{\mathrm{d}^d k}{(2\pi)^d} e^{-i(\tilde{E}_n(k)t - k\cdot x)} \left(\langle GS|j^0(0)|nk\rangle\langle nk|I(y)|GS\rangle - \text{c.c.}\right) \\
&= \sum_n \int \mathrm{d}^d k \delta_\Omega(k) \left(\langle GS|j^0(0)|nk\rangle\langle nk|I(y)|GS\rangle - \text{c.c.}\right),
\end{aligned} \tag{20}$$

where in the second line we have replaced the integral over spacetime by the integral over the spacetime region Ω, by assuming locality of the operators; we have also inserted a complete relation and shifted the charge operator. In the third line, $\tilde{E}_n(k) \equiv E_n(k) - E_{GS}$, and we have assumed translational invariance of the ground state and set its momentum to zero.

In the last line, a peak function $\delta_\Omega(k)$ near $k \approx 0$ is introduced by integrating x out. As a result of the non-vanishing expectation value, we see that there should be an excitation near the zero momentum such that

$$\begin{cases} I(y) \left|GS\right\rangle \sim \left|n, \delta q\right\rangle \\ j^0(0) \left|GS\right\rangle \sim \left|n, \delta q\right\rangle \end{cases}, \tag{21}$$

for some n. Notice that in the limit $\Omega \to \infty$, we shall have $\delta q \to 0$, which indicates a zero-wavevector excitation. To further illustrate the gapless nature, we utilize the charge conservation

$$\begin{aligned} 0 &= \partial_t \left\langle GS\right| \left[Q, I(y)\right] \left|GS\right\rangle \\ &= -i \sum_n \int \mathrm{d}^d k \delta_\Omega(k) \tilde{E}_n(k) \left(\left\langle GS|j^0(0)|nk\right\rangle \left\langle nk|I(y)|GS\right\rangle - \text{c.c.} \right), \end{aligned} \tag{22}$$

and since all terms except $\tilde{E}_n(k)$ contribute to the non-zero expectation value, we conclude that $\tilde{E}_n(\delta q \to 0) \to 0$ indicating the Goldstone mode as a gapless zero-momentum excitation. The counting of the Goldstone modes needs careful treatment, since both the Noether charge and the interpolating field operator can generate the *same* Goldstone mode, and when it is the case that the interpolating field operator is yet another symmetry generator, e.g., $[S_x, S_y] = iS_z$ in the ferromagnetic Heisenberg model, the Goldstone mode is not counted twice. Call this type of Goldstone mode as type-B, and the usual one as type-A, we have

$$\begin{cases} n_A = \dim(G/H) - \mathrm{rank}(M) \\ n_B = \mathrm{rank}(M)/2 \end{cases}, \tag{23}$$

where the Watanabe-Brauner matrix $M_{ab} = -i \left\langle GS|[Q_a, Q_b]|GS\right\rangle$ is homogeneous in space. It has been shown that in almost all cases type-A modes have linear dispersion while type-B modes have quadratic dispersion, with the latter understood from the coupling of two linear Bosonic modes as

$$\mathcal{L}_{eff} = 2M(\pi_1 \partial_t \pi_2 - \pi_2 \partial_t \pi_1) + \frac{1}{2}(\partial \pi_1)^2 + \frac{1}{2}(\partial \pi_2)^2, \tag{24}$$

whose equation of motion $[(\omega^2 - k^2)\sigma_0 - 2M\omega\sigma_y]\pi = 0$ leads to a quadratic excitation $\omega_- = \sqrt{k^2 + M^2} - M \approx \frac{k^2}{2M}$ and a massive mode.

One interesting point for spontaneous symmetry breaking lies in its correlation function over order parameter operator $\phi(x)$, which exhibits an off-diagonal long-range order as we

have discussed. However, as we have also discussed, any ground state can be chosen to evaluate the correlation function, without specifying the symmetry-broken or unbroken one. A more appropriate way to consider is to evaluate the connected correlation function, which is defined as

$$\langle\phi(x,t)\phi(x',t)\rangle_c = \langle\phi(x,t)\phi(x',t)\rangle - \langle\phi(x,t)\rangle\langle\phi(x',t)\rangle\,, \tag{25}$$

and reveals the long-range entanglement of the state. This quantity can be utilized as a signature of the spontaneous symmetry breaking, thanks to the short-range correlated nature of any symmetry-broken state, which shares a cluster decomposition in position. On the other hand, for an eigenstate of the symmetry operator, this quantity remains non-zero when $|x-x'|\to\infty$. This fact can be seen most illustratively through the ferromagnetic Ising model, which owns a $\mathbb{Z}_2$ symmetry by flipping all spins. The symmetry broken ground state reads $|\Uparrow\rangle = |\uparrow\uparrow\cdots\rangle$ and $|\Downarrow\rangle = |\downarrow\downarrow\cdots\rangle$, and the $\mathbb{Z}_2$ symmetric cat states are $|\pm\rangle = \frac{1}{\sqrt{2}}(|\Uparrow\rangle \pm |\Downarrow\rangle)$. For local order parameter σ_i^z, its expectation value reads $\langle\sigma_i^z\rangle = 0$ for $|\pm\rangle$, and $\langle\sigma_i^z\rangle = \pm 1$ for $|\Uparrow\rangle$ and $|\Downarrow\rangle$; the correlation function reads $\langle\sigma_i^z\sigma_j^z\rangle = 1$ for any state; while the connected correlation function reads $\langle\sigma_i^z\sigma_j^z\rangle_c = 1$ for $|\pm\rangle$, and $\langle\sigma_i^z\sigma_j^z\rangle_c = 1-1 = 0$ for $|\Uparrow\rangle$ and $|\Downarrow\rangle$, which confirms our statement. The cat states are unstable against any perturbation, under the thermodynamic limit or even for large-enough system. This motivates us for an equivalent definition of the spontaneous symmetry breaking, that the symmetric ground state can not be short-range correlated, vice versa.

B. Spontaneous breaking of discrete time translational symmetry

After making the concept clear, we can attempt to define a discrete time crystal. The discrete time translational symmetry of the Hamiltonian $H(t) = H(t+T)$ indicates that a symmetry unbroken state should be a Floquet eigenstate of the Floquet operator $U_f = U(T,0)$ with the same evolution period as $U_f\,|\psi(0)\rangle_{Sym} = |\psi(0)\rangle_{Sym}$ up to a phase, thus showing the oscillating behavior of an observable with the same period, too. Then the spontaneous symmetry breaking should be signaled by a subharmonic response, with the observable oscillating with a period mT with m an integer larger than one[8], i.e.,

$$\langle\psi(t+T)|\phi(x)|\psi(t+T)\rangle \neq \langle\psi(t)|\phi(x)|\psi(t)\rangle\,, \tag{26}$$

where $|\psi(t+T)\rangle = U(t+T,t)\,|\psi(t)\rangle$, with the time evolution operator $U(t+T,t) = \mathcal{T}\exp\left(-i\int_t^{t+T} H(t')\mathrm{d}t'\right)$. Equivalently, we can write $\langle\psi|\phi(x,t+T)|\psi\rangle \neq \langle\psi|\phi(x)|\psi\rangle$ in the Heisenberg picture. Counting on stability against long-time behavior, in the case that the dynamics exhibit m-fold time-translational symmetry breaking, we can write[5]

$$\lim_{\tau\to\infty}\frac{1}{\tau}\sum_{n=1}^{\tau}\langle\phi(x,(mn+p)T)\rangle \neq \lim_{\tau\to\infty}\frac{1}{\tau}\sum_{n=1}^{\tau}\langle\phi(x,(mn)T)\rangle\,, \tag{27}$$

where $0<p<m$ corresponds to the m phases of the orbit, with equality restored for $p=m$.

We can indeed formulate the language of connected correlation function for the DTC case similarly. Notice that in the Heisenberg picture, the definition of a DTC from Eq. (26) can be rewritten as $\langle\psi|\tilde{\phi}(x)|\psi\rangle \neq 0$, where $\tilde{\phi} = \phi(T) - \phi(0)$ serves as the more appropriate order parameter operator for the DTC. Since in the case the broken symmetry is a discrete one, generally we can assume an m-fold symmetry breaking case as mentioned, and we can define the corresponding short-range correlated subspace spanned by $|\lambda\rangle\,, \lambda = 0, \cdots, m-1$ as symmetry breaking states. Such a subspace should also be a subspace of the eigenstates of the Floquet operator U_f[6], i.e., $|\psi\rangle_{Sym} = \sum_\lambda c_\lambda\,|\lambda\rangle$, and the space is an eigenstate multiplet well isolated with other states. By a good-enough choice of the order parameter operator, the properties should be satisfied as:

• The different short-range correlated basis states are locally distinguishable, i.e., $\langle\tilde{\phi}(x)\rangle_\lambda \neq \langle\tilde{\phi}(x)\rangle_{\lambda'}$ for $\lambda \neq \lambda'$;

• The different short-range correlated basis states are not connectable by local operators, i.e., $\langle\lambda|A|\lambda'\rangle = 0$ for $\lambda \neq \lambda'$ with any local operator A;

• Any two different short-range correlated basis states can be related by some symmetry operator based on U_f, and U_f acts transitively on the short-range correlated basis states, i.e., $\langle U_f\rangle_\lambda = 0$.

We firstly focus on the two-point function $\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle$ with $|x-y|\to\infty$, which we claim that is non-zero for both the symmetric and symmetry-broken states. To see this, we firstly write

$$\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle_\lambda = \langle\tilde{\phi}\rangle_\lambda^2 \neq 0, \tag{28}$$

based on the short-range correlated and locally distinguishable nature, with assumed translational invariance of the basis states. Notice that the locality of the order parameter operator is crucial, and is protected by the Lieb-Robinson bound involving time evolution.

Then

$$\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle_{Sym}=\sum_{\lambda\lambda'}c_{\lambda}^{*}c_{\lambda'}\langle\lambda|\tilde{\phi}(x)\tilde{\phi}(y)|\lambda'\rangle=\sum_{\lambda}|c_{\lambda}|^{2}\langle\tilde{\phi}\rangle_{\lambda}^{2}\neq 0. \tag{29}$$

However, for the connected correlation function $\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle_{c}=\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle-\langle\tilde{\phi}(x)\rangle\langle\tilde{\phi}(y)\rangle$, we have $\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle_{c}^{\lambda}=0$ that follows directly from the locality, while

$$\langle\tilde{\phi}(x)\tilde{\phi}(y)\rangle_{c}^{Sym}=\sum_{\lambda}|c_{\lambda}|^{2}\langle\tilde{\phi}\rangle_{\lambda}^{2}-\left(\langle\tilde{\phi}\rangle_{Sym}\right)^{2}\neq 0, \tag{30}$$

where the symmetric nature of the state $|\psi\rangle_{Sym}$ makes the expectation value of the order parameter operator vanish. From the comparison, the long-range nature for the symmetric state is revealed. A similar derivation is also suitable for the static case.

Since a discrete time crystal involves non-trivial order in the time domain, a statement involving correlation change for the spontaneous symmetry breaking can be made, such that the symmetric eigenstates of U_f will satisfy

$$\langle\psi_{Sym}|\,\phi(x,T)\phi(y,0)\,|\psi\rangle_{Sym}-\langle\psi_{Sym}|\,\phi(x,0)\phi(y,0)\,|\psi\rangle_{Sym}\neq 0, \tag{31}$$

even as $|x-y|\to\infty$. To show this, we notice that by locality,

$$\begin{aligned}\langle\psi_{Sym}|\,\phi(x,T)\phi(y,0)\,|\psi\rangle_{Sym}&=\sum_{\lambda\lambda'}c_{\lambda}^{*}c_{\lambda'}\langle\lambda|\phi(x,T)\phi(y,0)|\lambda'\rangle\\&=\sum_{\lambda}|c_{\lambda}|^{2}\langle\lambda|\phi(x,T)\phi(y,0)|\lambda\rangle\\&=\sum_{\lambda}|c_{\lambda}|^{2}\langle\phi(x,T)\rangle_{\lambda}\langle\phi(y,0)\rangle_{\lambda},\end{aligned} \tag{32}$$

so does that $\langle\psi_{Sym}|\,\phi(x,0)\phi(y,0)\,|\psi\rangle_{Sym}=\sum_{\lambda}|c_{\lambda}|^{2}\langle\phi(x,0)\rangle_{\lambda}\langle\phi(y,0)\rangle_{\lambda}$. Then, the difference reads

$$\begin{aligned}&\langle\psi_{Sym}|\,\phi(x,T)\phi(y,0)\,|\psi\rangle_{Sym}-\langle\psi_{Sym}|\,\phi(x,0)\phi(y,0)\,|\psi\rangle_{Sym}\\&=\sum_{\lambda}|c_{\lambda}|^{2}\left(\langle\phi(x,T)\rangle_{\lambda}-\langle\phi(x,0)\rangle_{\lambda}\right)\langle\phi(y,0)\rangle_{\lambda}\\&=-\frac{1}{2}\sum_{\lambda}|c_{\lambda}|^{2}\left(\langle\phi(T)\rangle_{\lambda}-\langle\phi(0)\rangle_{\lambda}\right)^{2},\end{aligned} \tag{33}$$

where to arrive at the last line we have utilized translational invariance, together with the fact that $\sum_{\lambda}|c_{\lambda}|^{2}\langle\phi(T)\rangle_{\lambda}^{2}=\sum_{\lambda}|c_{\lambda}|^{2}\langle\phi(0)\rangle_{\lambda}^{2}$. This can be seen by viewing U_f as a symmetry transition between $|\lambda\rangle$s, thus $\sum_{\lambda}|c_{\lambda}|^{2}\langle\phi(T)\rangle_{\lambda}^{2}=\sum_{\lambda}|c_{\sigma(\lambda)}|^{2}\langle\phi(0)\rangle_{\lambda}^{2}$, and requiring $|\psi\rangle_{Sym}$ to be invariant under U_f leads to $|c_{\lambda}|=|\langle\lambda|\psi\rangle_{Sym}|=|\langle\lambda|U_f|\psi\rangle_{Sym}|=|c_{\sigma(\lambda)}|$, thus proving the equality. Given Eq. (33), we can see that the non-zero correlation change is equivalent to the requirement of spontaneous symmetry breaking.

C. Little introduction to the many-body localized discrete time crystal

Many-body localization (MBL)[10] is a phenomenon that is believed to occur in 1D systems with strong quenched disorder and short-range interactions. It represents a failure of the eigenstate thermalization hypothesis (ETH) with ergodicity breaking, and is characterized by the emergence of an extensive set of quasi-local integrals of motion (LIOMs), i.e. emergent symmetry operators. It is this emergent integrability that strongly constrains the system's dynamics and thus provides an intuitive explanation of why it fails to thermalize. Thus, MBL systems provide a fertile ground for the realization of stabilized DTCs.

In [11] and [12], people both considered a two-step Hamiltonian for spin-1/2 systems in a single Floquet period, with one step a MBL Hamiltonian H_{MBL} and the other step a spin-flipping H_{flip}. The idea is straightforward: the MBL Hamiltonian provides the necessary integrability to stabilize the DTC, while the spin-flip Hamiltonian breaks the discrete time-translational symmetry, by splitting a precession motion into two (or more) periods. However, the writer thinks that people has given some misleading statements, since they write the Floquet operator as $U_f = \exp(-iH_{MBL}t_1)\exp(-iH_{filp}t_2)$, which is incorrect considering the non-commutating nature of the two parts. This comes from the singular behavior of the Hamiltonian between two steps. Nevertheless, by assuming that there exists such a Hamiltonian that allows for such a Floquet operator, the DTC can be stabilized as shown in Fig. 1,2. Especially in the latter, the essence of the discrete time crystal is revealed: despite imperfect spin rotations, collective synchronization from the interactions maintains robust oscillations at half the driving frequency.

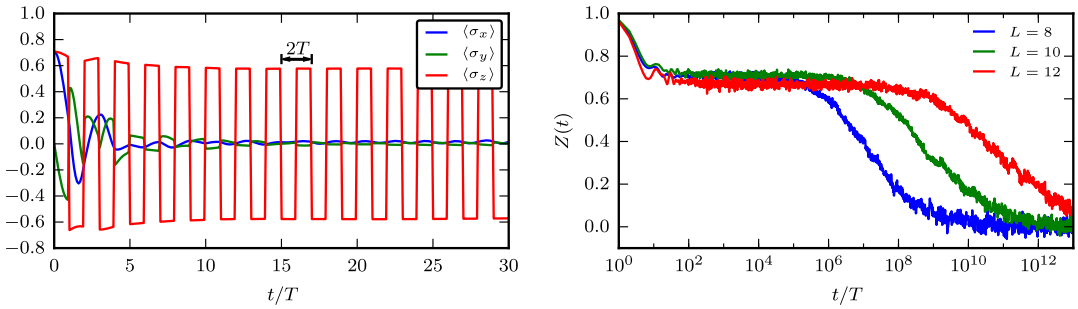


FIG. 1: $Z(t) = \overline{(-1)^t \langle \sigma_i^z(t) \rangle}$. Cut from Ref. [11].

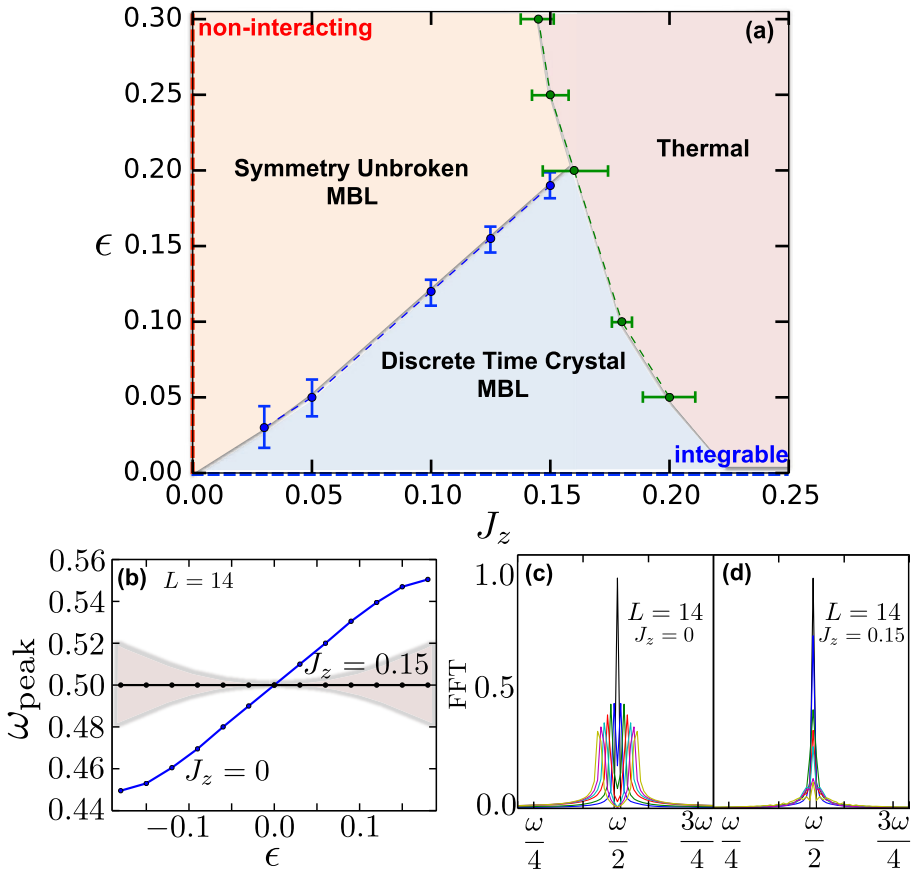


FIG. 2: (a) Phase diagram of the discrete time crystal as a function of interaction strength J_z and pulse imperfections ϵ. (b) Depicts the location of the subharmonic Fourier peak as a function of ϵ. In the noninteracting case ($J_z = 0$), the peak tracks ϵ, while in the interacting case ($J_z = 0.15$), the peak remains rigidly locked at $\omega/2$. The pink region indicates the full width at half maximum (FWHM) of the base of the $\omega/2$ peak. Data are obtained at $L = 14$ with 10^2 disorder averages. (c)-(d) Representative realizations of the subharmonic Fourier response corresponding to ϵ in (b). All Fourier transforms are computed using $10 < n < 150$ Floquet periods. Cut from Ref. [12].

D. Little introduction to the prethermal discrete time crystal

It was argued that under driving, generic many-body systems that obey the eigenstate thermalization hypothesis, eventually heat to up to an infinite temperature, featureless state. However, it has been shown[9] that, if the frequency of the drive ω_D is very large compared to the local energy scales J of the system, a general bound for the linear-response heating

rates exists suppressed exponentially by the driving frequency, and there is a long-lived quasi-steady state called a prethermal state. The proof again relies on the locality of the system and the Lieb-Robinson bound. The physical picture behind such type of DTC can be stated as: in systems with disparate energy scales, the dynamics (and thus thermalization) of "fast moving" degrees of freedom are governed by an effective Hamiltonian where the slow degrees of freedom remain basically frozen, since the system must undergo $\sim \omega_D/J$ local rearrangements in order to absorb a single unit of energy from the drive. Then the spontaneously breaking of the emergent symmetry of H_{eff} leads to the establishment of a DTC.

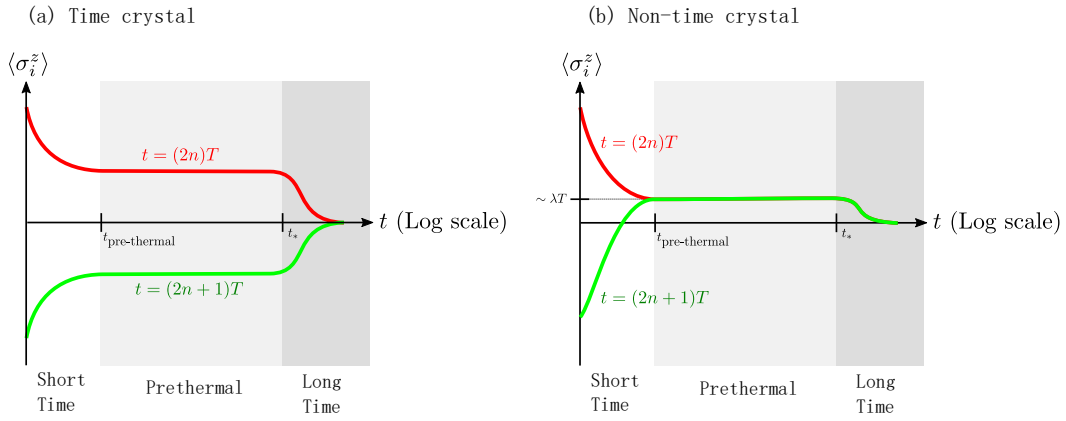


FIG. 3: Schematic depicting prethermal time-crystalline order identified via the stroboscopic behavior of the magnetization as a function of time. Left panel: in the prethermal discrete time-crystal phase, the magnetization oscillates throughout the prethermal regime, leading to stroboscopic curves that exhibit plateau behavior. The prethermal time crystal ultimately melts at exponentially late times, controlled by the frequency of the driving field. Right panel: in the non-time-crystal prethermal phase, the magnetization during even and odd Floquet cycles becomes identical after a short time (corresponding to local thermalization). During the intermediate prethermal regime, this magnetization may remain small, but nonzero, until it decays to zero at the Floquet heating timescale. Cut from Ref. [5, 13].

In [13], people prove that for $H(t) = H_0(t) + V(t)$, with the local energy scale of V the small one, satisfying the large frequency condition with $X^N = 1$, where $X = U_0(T, 0)$, there

exists a time-independent unitary transformation such that

$$\mathcal{U}U_f\mathcal{U}^\dagger = Xe^{iDT} + O(e^{-\omega_D/NJ}), \tag{34}$$

where D is a local time-independent Hamiltonian satisfying $[D, X] = 0$. The effective Hamiltonian $H_{eff} = \mathcal{U}^\dagger D\mathcal{U}$ shares a hidden $\mathbb{Z}_N$ symmetry generated by $\mathcal{U}^\dagger X\mathcal{U}$. And spontaneous breaking of the $\mathbb{Z}_N$ symmetry leads to the prethermal DTC. For instance, consider the $N = 2$ case with $X = \prod_i \sigma_i^x$, and $D = -J\sum_{\langle i,j\rangle} \sigma_i^z\sigma_j^z$, then for the order parameter σ_i^z, a time-crystalline behavior emerges since

$$\begin{aligned} M_{2n+1} &= \langle\tilde{\psi}((2n+1)T)|\sigma_i^z|\tilde{\psi}((2n+1)T)\rangle \\ &= \langle\tilde{\psi}(2nT)|\, e^{-iDT}X^\dagger\sigma_i^z Xe^{iDT}\,|\tilde{\psi}(2nT)\rangle \\ &= -\,\langle\tilde{\psi}(2nT)|\,\sigma_i^z\,|\tilde{\psi}(2nT)\rangle = -M_{2n}, \end{aligned} \tag{35}$$

where $|\tilde{\psi}\rangle = \mathcal{U}\,|\psi\rangle$. An illustrative picture of the example is shown in Fig. 3.

In Hu's paper[14], the concept of prethermal DTC is further clarified by considering dynamical symmetry in the time domain, like the crystalline symmetry in the usual lattice system. Specifically, the Hamiltonian $H(t) = H_0(t) + V$ where $V = J\sum_i \sigma_i^\mu\sigma_{i+1}^\mu$ is the short-range many-body interaction, and $H_0(t) = \sum_i \frac{1}{2}\hbar\Omega\sin(\omega t)\sigma_i^x + \hbar\Omega\sin^2(\omega t/2)\sigma_i^y$ is the non-interacting part, which plays the role of spin-flipping. $H_0(t)$ shares a non-symmorphic gliding symmetry in time and forces the system to contain Möbius twisted instantaneous eigenstates when the parameter $\alpha = 8\Omega/\omega$ is fine-tuned. When α deviates from the perfect value, the system will heat up. However, strikingly, the prethermal DTC can be stabilized by introducing long-range interactions, as shown in Fig. 4. The result is significant since it directly illustrates that interactions can stabilize the prethermal DTC even in 1D, from the situation where the making-system-fully-periodic parameter deviates.

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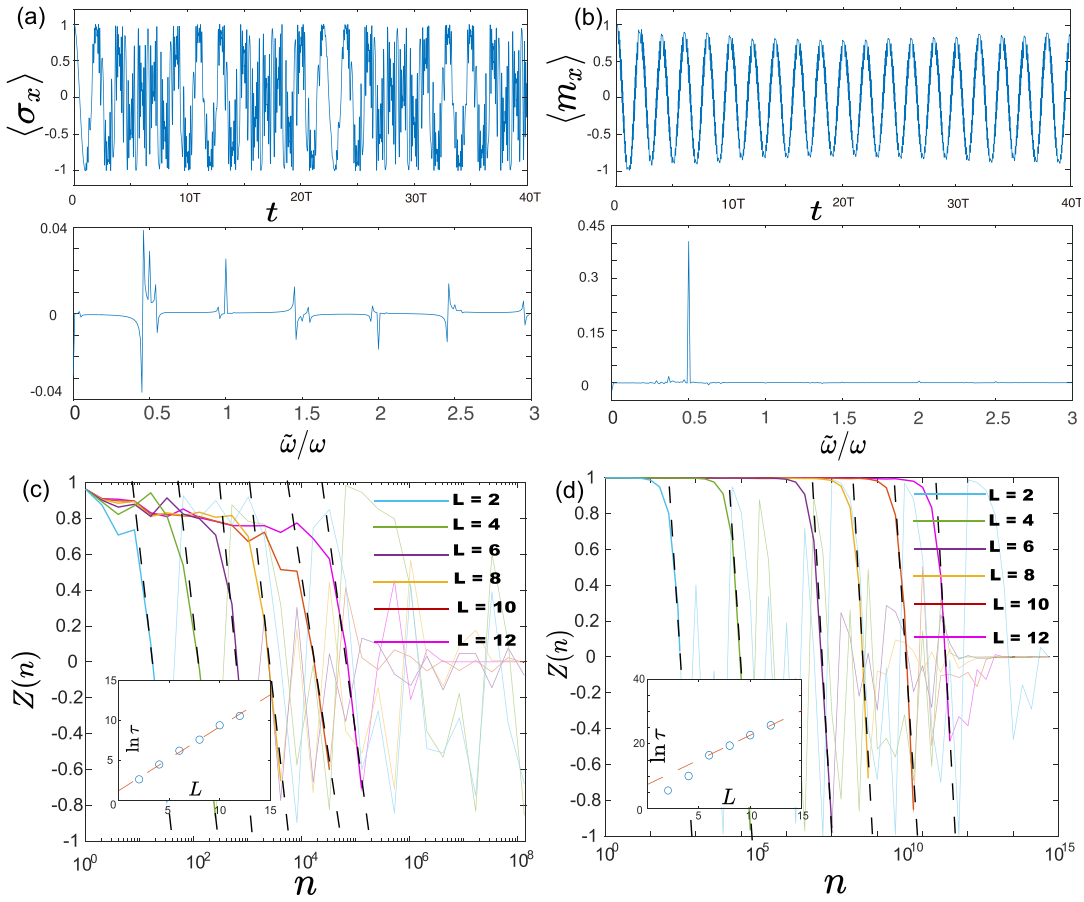


FIG. 4: (a) $\langle\sigma_x(t)\rangle$ and its Fourier spectrum for noninteracting system for the first 40 periods. (b) $\langle m_x(t)\rangle$ (spatially averaged $\langle\sigma_i^x(t)\rangle$) and its Fourier spectrum for the interacting system in the x direction with an initial state polarized in the x direction with length $L = 10$. (c) $Z(n) = (-1)^n \langle m_x(t)\rangle$ at the stroboscopic time, with different length chains showed in the legend. The inset shows the lifetime via the size L of the system. $\tau \propto \exp(bL)$, with $b = 1.56$ for $\alpha = 81.60$ (slightly deviated from one perfect period $\alpha_{13} = 80.07$). (d) $Z(n)$ at the stroboscopic time for $\alpha = \alpha_{13}$, $b = 2.49$. Here we set the interaction $J = 0.2\hbar\omega$. Cut from Ref. [14].

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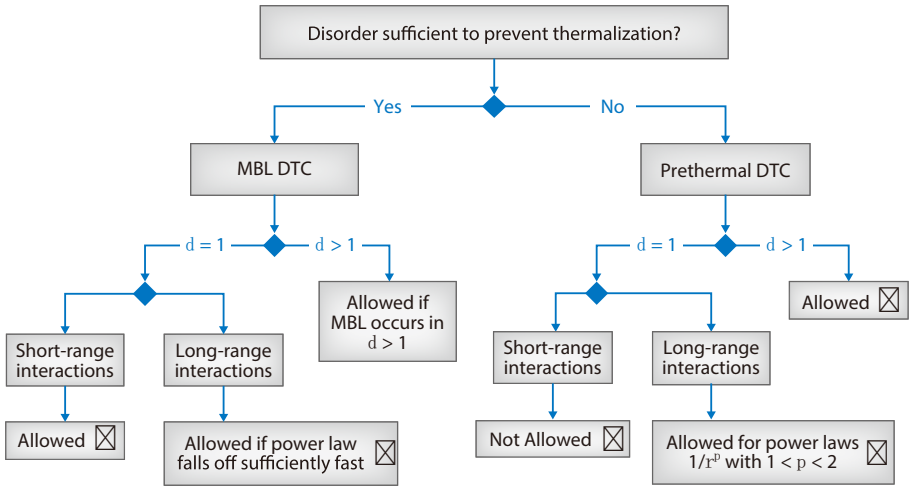


FIG. 5: Flowchart clarifying the key differences between the MBL and prethermal DTC. Especially striking is the following dichotomy: The MBL DTC is well established in $d = 1$ for systems with short-range interactions, whereas long-range interactions tend to destabilize localization. However, the prethermal DTC is forbidden in $d = 1$ with short-range interactions, but can be stabilized in the presence of long-range interactions. At its core, this fact owes to the absence of finite temperature Ising symmetry breaking in $d = 1$ with short-range interactions. Furthermore, this difference again highlights the fact that MBL eigenstates behave as thermal states with zero temperature, whereas the prethermal eigenstates of HF behave as finite temperature thermal states. Abbreviations: DTC, discrete time crystal; MBL, many-body localization. Cut from Ref. [6].

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